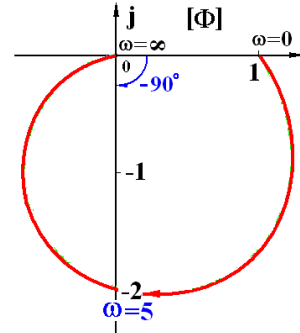


4. 单位反馈的典型欠阻尼二阶系统，校正前
其闭环幅相特性如图所示。

- (1) 求校正前系统的截止频率 ω_{c0} ，相角裕度 γ_0 及动态指标 ($\sigma\%$, t_s)；

- (2) 若采用串联校正装置 $G_c(s) = \frac{s}{\frac{s}{2.5} + 1}$ ，

求校正后系统的 ω_c ， γ ， $\sigma\%$ 和 t_s 。



解. 依图可设校正前闭环传递函数

$$\Phi_0(s) = \frac{1}{\left(\frac{s}{\omega_{n0}}\right)^2 + 2\xi_0 \frac{s}{\omega_{n0}} + 1}$$

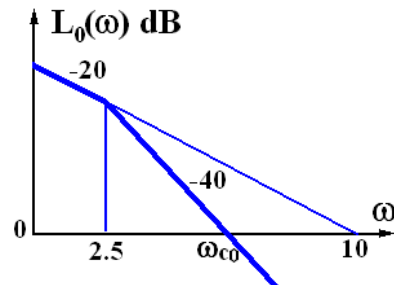
$$\angle \Phi_0(j\omega) = \arctan \frac{2\xi_0 \frac{\omega}{\omega_{n0}}}{1 - \frac{\omega^2}{\omega_{n0}^2}} \stackrel{\omega=5}{=} -90^\circ \quad \Rightarrow \quad \omega = \omega_{n0} = 5$$

$$|\Phi_0(j\omega)| = \frac{1}{\sqrt{\left(1 - \frac{\omega^2}{\omega_{n0}^2}\right)^2 + \left(2\xi_0 \frac{\omega}{\omega_{n0}}\right)^2}} \stackrel{\omega=10=\omega_0}{=} \frac{1}{2\xi_0} = 2 \quad \Rightarrow \quad \xi_0 = \frac{1}{4}$$

$$\Phi_0(s) = \frac{1}{\left(\frac{s}{5}\right)^2 + 2 \times \frac{1}{4} \times \frac{s}{5} + 1} = \frac{25}{s^2 + 2.5s + 25}$$

$$G_0(s) = \frac{\Phi_0(s)}{1 - \Phi_0(s)} = \frac{10}{s\left(\frac{s}{2.5} + 1\right)}$$

$$\omega_{c0} = \sqrt{2.5 \times 10} = 5$$



$$\gamma_0 = 180^\circ - 90^\circ - \arctan \frac{5}{2.5} = 90^\circ - 63.4^\circ = 26.6^\circ$$

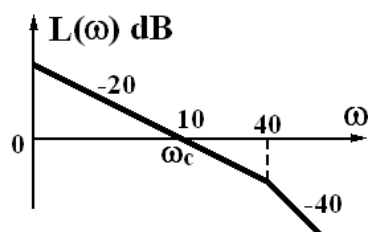
$$\begin{cases} \omega_{n0} = \sqrt{25} = 5 \\ \xi_0 = 0.25 \end{cases} \quad \begin{cases} \sigma \% = e^{-\xi_0 \pi / \sqrt{1-\xi_0^2}} = 44.4 \% \\ t_s = \frac{3.5}{\xi_0 \omega_{n0}} = \frac{3.5}{1.25} = 2.8 \end{cases}$$

(2) 校正后

$$G(s) = G_c(s) \cdot G_0(s) = \frac{10(\frac{s}{2.5} + 1)}{s(\frac{s}{2.5} + 1)(\frac{s}{40} + 1)}$$

$$= \frac{10}{s(\frac{s}{40} + 1)} = \frac{400}{s(s + 40)}$$

$$\omega_c = 10$$



$$\gamma = 180^\circ - 90^\circ - \arctan \frac{10}{40} = 90^\circ - 14^\circ = 76^\circ$$

$$\Phi(s) = \frac{400}{s^2 + 40s + 400} \quad \begin{cases} \omega_n = \sqrt{400} = 20 \\ \xi = \frac{40}{2 \times 20} = 1 \end{cases}$$

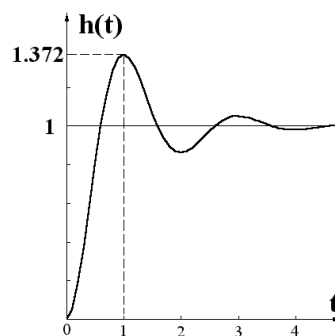
$$\begin{cases} \sigma \% = 0 \\ t_s = \frac{4.75}{T_1} = \frac{4.75}{\omega_n} = \frac{4.75}{20} = 0.2375 \end{cases}$$

5. 某单位反馈的典型二阶系统，其单位阶跃响应如图所示。

(1) 系统的开环传递函数，画出系统的结构图；

(2) 用适当的校正方式，并调整开环增益，使校正后系统为超调量 $\sigma \% = 16.3 \%$

调节时间 $t_s = 1$ 秒的欠阻尼二阶系统，试画出校正后系统的结构图，确定校正装置的传递函数和系统的开环增益。

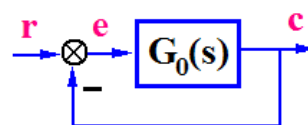


系统单位阶跃响应

解. (1) 依题意, 校正前系统动态指标为

$$\begin{cases} \sigma_0 \% = e^{-\xi_0 \pi / \sqrt{1-\xi_0^2}} = 37.2 \% \\ t_{0p} = \pi / \sqrt{1-\xi_0^2} \omega_{n0} = 1 \end{cases}$$

联立求解 $\begin{cases} \xi_0 = 0.3 \\ \omega_{n0} = 3.3 \end{cases}$



$$\Phi_0(s) = \frac{\omega_{n0}^2}{s^2 + 2\xi_0 \omega_{n0} s + \omega_{n0}^2} = \frac{3.3^2}{s^2 + 1.98s + 3.3^2}$$

$$G_0(s) = \frac{\Phi_0(s)}{1 - \Phi_0(s)} = \frac{3.3^2}{s(s + 1.98)} = \frac{10.89}{s(s + 1.98)} = \frac{5.5}{s(\frac{s}{1.98} + 1)}$$

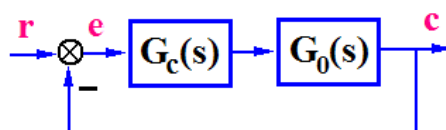
解法一 用串联校正

依题意, 校正后闭环传递函数应为

$$\begin{aligned} \Phi(s) &= \frac{\omega_n^2}{s^2 + 2\xi \omega_n s + \omega_n^2} \\ &\downarrow \begin{cases} \sigma \% = 16.3 \% \\ t_s = 3.5 / \xi \omega_n = 1 \end{cases} \quad \begin{cases} \xi = 0.5 \\ \omega_n = 7 \end{cases} \\ &= \frac{7^2}{s^2 + 7s + 7^2} = \frac{49}{s^2 + 7s + 49} \end{aligned}$$

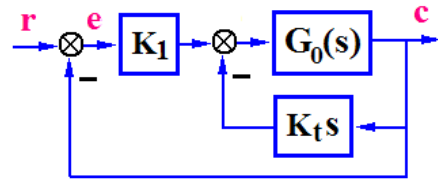
$$G(s) = \frac{\Phi(s)}{1 - \Phi(s)} = \frac{49}{s(s + 7)} = \frac{7}{s(\frac{s}{7} + 1)} = G_c(s) \cdot G_0(s) \quad \begin{cases} K = 7 \\ v = 1 \end{cases}$$

$$G_c(s) = \frac{G(s)}{G_0(s)} = \frac{7}{s(\frac{s}{7} + 1)} \cdot \frac{s(\frac{s}{1.98} + 1)}{5.5} = 1.273 \times \frac{\frac{s}{1.98} + 1}{\frac{s}{7} + 1}$$



解法二 用测速反馈

$$G_0(s) = \frac{\Phi_0(s)}{1 - \Phi_0(s)} = \frac{10.89}{s(s+1.98)} = \frac{5.5}{s(\frac{s}{1.98} + 1)}$$



$$G(s) = K_1 \frac{10.89}{s(s+1.98)} = \frac{10.89 K_1}{s(s+1.98 + 10.89 K_t)}$$

$$\Phi(s) = \frac{G(s)}{1 + G(s)} = \frac{10.89 K_1}{s^2 + (1.98 + 10.89 K_t)s + 10.89 K_1} = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$\text{令} \begin{cases} t_s = \frac{3.5}{\xi\omega_n} = 1 \\ \sigma\% = 16.3 \end{cases} \quad \begin{cases} \xi\omega_n = 3.5 = \frac{1.98 + 10.89 K_t}{2} \\ \xi = 0.5 = \frac{3.5}{\sqrt{10.89 K_1}} \end{cases} \quad \begin{cases} K_t = 0.461 \\ K_1 = 4.5 \end{cases}$$